# Naomi R. Shvedov, PhD

*Postdoctoral Associate · Sorrells Lab · University of Pittsburgh*

naomi.shvedov@pitt.edu · naomishved.github.io · ORCID: 0000-0001-5550-5231

## APPOINTMENTS

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| 2025 – present | **Postdoctoral Associate**, Sorrells Lab, Department of Neuroscience, University of Pittsburgh |

## EDUCATION

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| 2019 – 2025 | **Ph.D., Neuroscience**, Boston University<br>Dissertation: Dynamics and mechanisms of neuron migration in the adult songbird brain.<br>Advisor: Benjamin B. Scott. |
| 2015 – 2019 | **B.S., Cell Biology and Neuroscience**, Rutgers University<br>Graduated Magna cum laude. Rutgers Cell Biology and Neuroscience Honors Program. |

## HONORS, AWARDS, AND FELLOWSHIPS

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| 2026 | Center for the Neural Basis of Cognition (CNBC) Professional Development Award |
| 2022 | Chan Zuckerberg Initiative (CZI) Bioimaging North America Program Professional Development Award |
| 2020 – 2022 | NSF NRT Neurophotonics Fellowship, Boston University |

## PUBLICATIONS

***Peer-reviewed:***

**Shvedov, N.R.**, Castonguay, S.J., Rother, A., Schick, D., Kornfeld, J., Scott, B. (2026). Songbird connectome reveals tunneling of migratory neurons in the adult striatum. *Current Biology*, vol. 36, issue 9, pp. 2242–2254.e3. https://doi.org/10.1016/j.cub.2026.03.057

**Shvedov, N.R.**, Analoui, S., Dafalias, T., Bedell, B.L., Gardner, T.J., Scott, B.B. (2024). In vivo imaging in transgenic songbirds reveals superdiffusive neuron migration in the adult brain. *Cell Reports*. vol. 43, issue 2, 113759. https://doi.org/10.1016/j.celrep.2024.113759

Mamounis, K.J., **Shvedov, N.R.**, Margolies, N., Yasrebi, A., Roepke, T.A. (2019). The effects of dietary fatty acids in the physiological outcomes of maternal high-fat diet on offspring energy homeostasis in mice. *Journal of Developmental Origins of Health and Disease*. pp. 1–12. doi:10.1017/s2040174419000540

***Manuscripts in preparation:***

Frostig, H., **Shvedov, N.R.**, Scott, B.B. (2025). Deep brain three-photon imaging in songbirds. *In preparation*.

***Spotlights:***

**Shvedov, N.R.**, Sorrells, S.F. (2025). Why reinvent the wheel? An idea to watch: extending growth cones to migrating neurons. *BioEssays*. https://doi.org/10.1002/bies.70059

**Shvedov, N.R.**, Margolis, D.J. (2019). An optical exposé of cortical function. *Trends in Neurosciences*. vol. 42, no. 8, pp. 511–513. doi:10.1016/j.tins.2019.05.003

## INVITED TALKS

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| Jun 2024 | CoNNExINs Symposium, New York University |
| Feb 2024 | Neurophotonics Trainee Tech Talk, Boston University |
| May 2023 | Neurodevelopment for Experiment–Theory Interaction Workshop, Harvard University |
| Mar 2023 | Cosyne Workshop on Neural Development and Evolution, Mont Tremblant, QC |
| Aug 2022 | Gordon Research Conference on Neural Development, Salve Regina University |

## CONFERENCE PRESENTATIONS

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| Apr 2026 | **Shvedov, N.R.**, Bandi, A.C., Runyan, C.A., Sorrells, S.F. *Uncovering the role of a "late bloomer" — the paralaminar amygdala nucleus.* University of Pittsburgh Postdoctoral Association Symposium. |
| Oct 2024 | **Shvedov, N.R.**, Castonguay, S.J., Rother, A., Schick, D., Kornfeld, J., Scott, B. *EM connectomics reveal tunneling of migratory neurons in adult songbird brain.* Janelia Conference for Developmental Specification of Complex Behaviors. |
| Nov 2023 | **Shvedov, N.R.**, Frostig, H., Mertz, J., Scott, B.B. *Deep brain three-photon imaging in transgenic songbirds.* Society for Neuroscience Annual Meeting. |
| 2022 | **Shvedov, N.R.**, Gardner, T.J., Scott, B.B. *In vivo imaging in transgenic songbirds reveals superdiffusive neuron migration in the adult brain.* Presented at: Gordon Research Conference on Neural Development; Gordon Research Conference on Neural Mechanisms of Acoustic Communication; Society for Neuroscience Annual Meeting; UC Irvine CNCM Conference on the Structure, Function and Development of Neural Circuits. |

## SOFTWARE TOOLS

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| **TopoSlice** | Python; work in progress. 3D reconstruction from low resolution, serial histological sections, including the path, angle, and imaging field of view of implanted microendoscopes. |
| **ChairPile** | Work in progress. Turns written or spoken notes into organized project tasks and daily plans. |
| **Songbird migration** | MATLAB. Tracking, collective-migration, and simulation analyses for Shvedov et al. 2024 (github.com/naomishved/Transgenic-songbird-neuron-migration). |

## OTHER POSITIONS AND TRAINING

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| Dec 2022 | 1st International Mini2P Workshop, Moser Group, Kavli Institute, Trondheim |
| Jun 2022 | Frontiers in Neurophotonics Summer School, Université Laval |
| 2018 – 2019 | **Undergraduate Research Assistant**, Margolis Lab, Department of Cell Biology and Neuroscience, Rutgers University<br>*Developed GRIN lens implant surgeries and two-photon calcium imaging of striatal medium spiny neurons in head-fixed mice during sensory discrimination. Designed and 3D-printed head-fixation hardware. Trained graduate students.* |
| 2017 – 2018 | **Undergraduate Research Assistant**, Roepke Lab, Department of Animal Sciences, Rutgers University |

*Performed RNA extraction, reverse transcription, and qPCR to assess epigenetic influences of maternal high-fat diet on offspring energy homeostasis.*

## TEACHING AND MENTORING

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| 2024 – 2025 | **Teaching Fellow**, Introduction to MATLAB (NE/PS 212), Boston University. Led discussion sections, review sessions, and hands-on labs; taught four in-class lectures. |
| Fall 2023 | **Teaching Fellow**, Principles of Neuroscience (GMS NE 700), Boston University. Supported first-year PhD students. |
| Summer 2022 | **GROW Mentor**, Greater Boston Area Research Opportunities for Young Women. Mentored a high-school student in a STEM research project as part of a program supporting underrepresented students. |

## SERVICE AND OUTREACH

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| Feb 2026 | **Neuroscience Proseminar Lecturer**, University of Pittsburgh. |
| Nov 2025 | **Abstract Reviewer**, Cosyne 2026. |
| Jan – Feb 2024 | **Graduate Student Member**, Biology Faculty Search Committee (Neuroethology), Boston University. |
| 2020 – 2022 | **Member**, NSF NRT Neurophotonics Training Program Seminar Planning Committee, Boston University. Coordinated faculty seminars on state-of-the-art neurophotonics. |
| 2020 – 2021 | **Recruitment Co-Chair**, Neuroscience Graduate Student Organization, Boston University. Led virtual recruitment for the Graduate Program of Neuroscience. |
| Oct 2020 | **Mentor Judge & GPN Exhibitor**, SACNAS National Diversity in STEM Conference. |

## TECHNICAL SKILLS

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| **In vivo** | Chronic two- and three-photon calcium imaging through cranial windows, GRIN lenses, and microprisms in live mice and songbirds; head-fixed behavior; stereotaxic and neonatal viral injection; chronic GRIN lens implantation in deep brain (amygdala); chronic chemogenetic manipulation. |
| **Tissue & histology** | Ex vivo organotypic slice culture (mouse and human); immunohistochemistry; AAV transduction, confocal and epifluorescence imaging. |
| **Molecular** | RNA extraction, PCR, qPCR. |
| **Analysis** | MATLAB; Python; FIJI/ImageJ; custom analytical pipelines for calcium imaging, cell tracking, and 3D histology reconstruction. |
| **Fabrication** | 3D modeling and printing of custom behavioral, head-fixation, and implant hardware. |